

Information Processing 1 — Week 9 exercises

Complete each exercise on your own laptop before the end of the exercise class. When you finish an exercise, **immediately** ask an instructor to check your answer. Answers checked after the end of the class will have a 50% penalty.

Do not struggle for more than a few minutes with anything you cannot solve. If you are stuck or have any questions then **raise your hand** and an instructor will help you.

1 [2 points] Analysing words, reading input from a text file

Verify that you understand how to check a word for the presence or absence of a given letter, and that you can read a sequence of words from a text file. If you have any difficulty finishing this part quickly, ask for help.

1.1 Testing whether a word excludes certain letters

Write a function called `avoids(word, letters)` that returns `True` if `word` does not contain any of the `letters`. For example, `avoids("cat", "ct")` returns `False` but `avoids("dog", "ct")` returns `True`.

```
print(avoids("cat", "ct")) # False
print(avoids("dog", "ct")) # True
print(avoids("chimpanzee", "ct")) # False
print(avoids("elephant", "ct")) # False
print(avoids("mongoose", "ct")) # True
```

Hint: One way to look at this is that the `avoids()` should return `False` as soon as it detects that any one of the `letters` occurs in `word`, and only if there are none then it finally returns `True`.

1.2 Reading input from the user and from a file

Use your `avoids(word, letters)` function to write a program that prompts the user to enter a string of 'forbidden letters' and then prints the number of words in the file `words.txt` that do *not* contain any of those letters.

Can you find which combination of 5 letters excludes the *smallest* number of words from being counted?

▷▷ Ask an instructor to check and record your work **now** (before you run out of time).

2 [4 points] Selecting words from a file based on various criteria

Practice on a few more example problems that analyse words read from a file.

2.1 Testing whether a word includes certain letters

Write a function `uses_only(word, letters)` that returns `True` if `word` contains *only* the given `letters`.

```
print(uses_only("appeal", "apple")) # True
print(uses_only("apple", "apple")) # True
print(uses_only("apples", "apple")) # False
print(uses_only("lapel", "apple")) # True
print(uses_only("palpable", "apple")) # False
print(uses_only("palpable", "abple")) # True
```

Hint: One way to look at this is to return `False` if `word` contains any letter that is not in `letters`.

2.2 Selecting words that include only certain letters

Use your function `uses_only(word, letters)` to write a program that prints all the words in the file `words.txt` that use *only* these letters: `"acefhlo"`

2.3 Selecting words that include all given letters

Write a function called `uses_all(word, letters)`, that returns `True` if `word` uses every letter in `letters` at least once.

(It is possible to write `uses_all()` with only one line of code in the body of the function.)

2.4 Selecting words that contain all given letters

Use your function `uses_all(word, letters)` to write a program that counts how many words in `words.txt` use all of these letters: `"aeiou"`.

How many words use all the vowels `"aeiou"`?

How many words use all the letters `"aeiouy"`?

▷▷ Ask an instructor to check and record your work **now** (before you run out of time).

3 [2 points] Performing more difficult analyses of words

Try to solve two slightly more difficult problems.

3.1 Counting monotonic words

Write a function `is_monotonic(word)` that returns `True` if the letters in `word` are in alphabetical order from left to right (double letters are OK). For example, `is_monotonic("beefily")` returns `True` but `is_monotonic("beefiness")` returns `False` (because 'e' comes before 'n' in the alphabet).

How many monotonic words are there in the file `words.txt`?

3.2 Double letters

A double letter is one which occurs twice, such as 'oo' in 'hoof'. Consecutive double letters are two pairs of double letters that occur with no other letter in between them. For example, consecutive double letters 'tt' and 'ee' occur in the word 'committee', but 'mm' and 'tt' are not consecutive because the letter 'i' comes between them.

Write a program to detect and print all the words in the file `words.txt` which have *three consecutive* double letters.

Hints: You might benefit from a function `is_double(word, position)` that checks for the same letter in `word` at indexes `position` and `position+1`. For each word you could then easily check for three consecutive letters from index 0 up to (and including) the element at index `len(word)-6`.

Which words in the file contain triple double letters?

▷▷ Ask an instructor to check and record your work **now** (before you run out of time).

4 [2 points] Presenting information visually

A program that prints only text output can display information visually, for example by printing a histogram of results made from lines of asterisks of different lengths.

4.1 Average length of words

Write a program that calculates the average length of the words in the file `words.txt`. Print the result with one decimal place of precision (one digit after the decimal point).

Hint: The function `round(f, n)` rounds the floating point number `f` to `n` decimal places.

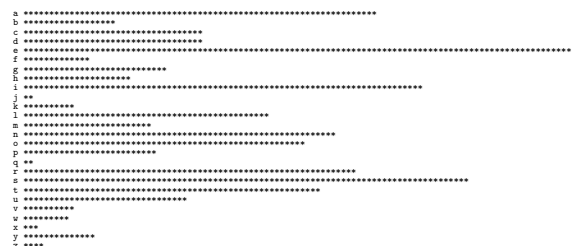
What is the average length of the words?

4.2 Printing histograms

The following program prints a *histogram* of how many times each letter of the alphabet appears in the `words.txt` file.

```
for letter in "abcdefghijklmnopqrstuvwxyz":
    words = open("words.txt")
    count = 0
    for line in words:
        word = line.strip()
        count = count + word.count(letter)
    if count > 0:
        # ensure at least one '*' is printed
        count = count + 1000
    print(letter, "*" * (count // 1000))
```

letter frequencies



What is the most common letter in the file?

Modify the program so that it prints a histogram of the *lengths* of the words in `words.txt`. For example, print one line of asterisks for each of the possible word lengths between 1 and 25 letters, inclusive.

What is the most common word length?

▷▷ Ask an instructor to check and record your work **now** (before you run out of time).

5 Challenges [1 bonus point]

Complete all the challenges before the end of the exercises class to earn a bonus point.

5.1 Counting words in a text file

The `readline()` function can be used for more than reading words. It can also read an entire line of text from a file.

Write a program that reads the text file `text.txt` containing English sentences and counts the number of words in the entire file. Note that each individual line might contain zero, one, or many words.

Hint: One way to approach this is to inspect each character in the line from left to right, looking for *transitions* from non-letters to letters. At the start of the line you are in 'non-letter' mode. When you see a letter you change mode from 'non-letter' to 'letter' and increment the word counter. Continue scanning and if you see a non-letter then you switch back to 'non-letter' mode.

5.2 Counting lines, words, and characters

Extend your program from the previous question to count the number of lines, words, and characters in `text.txt` at the same time. At the end of the program, print all three counts. (If you have MacOS or a Linux distribution then you can compare the output of your program with that of the `wc` program.)

▷▷ Ask an instructor to check and record your work **now** (before you run out of time).